L4 Safety Island — what changes in Autoware

Five phases, from the Autoware node diagram to the Safety Island alone

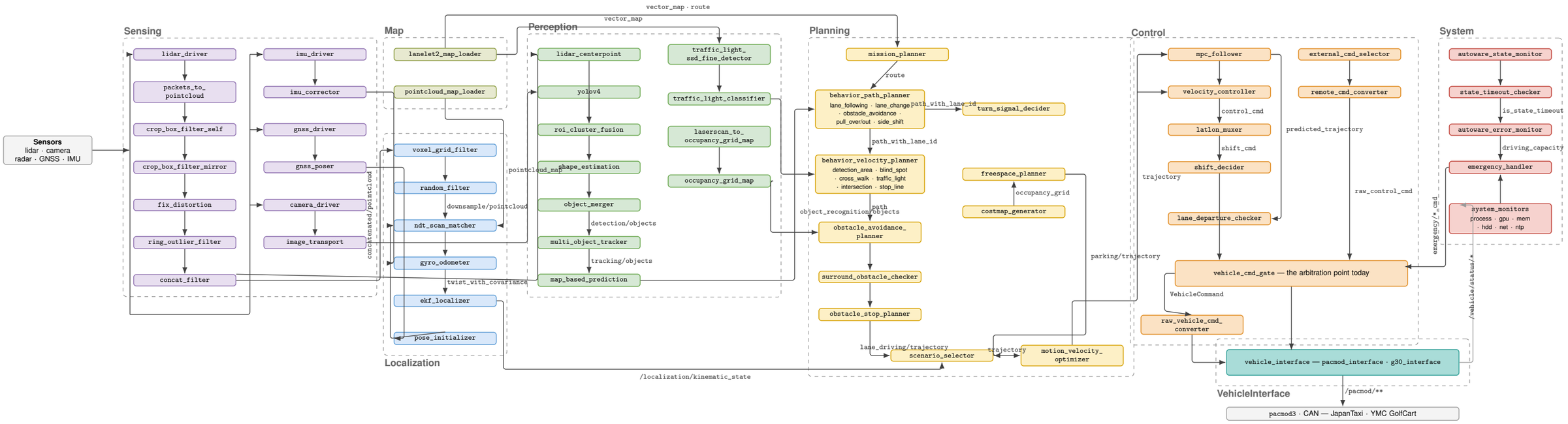
Phase 1	Autoware today, every node	The Autoware node graph as it stands today, redrawn in the series' own visual language so the five pages read as one progression
Phase 2	Autoware today, one box per stack	Every node of phase 1 merged into its stack, and every link merged with it
Phase 3	The same picture, after the island	Same boxes, same colours. **The Control stack and the Vehicle Interface migrate to the island**; the island grows reduced versions of the other stacks
Phase 4	What changed, at node level	A *selective* zoom: on the HPC only the nodes removed, re-sourced or newly designed; on the island only the nodes migrated from it. Everything else stays at the phase-3 level
Phase 5	The Safety Island alone	The HPC dropped entirely, so all 37 island nodes have room to be read
B1–B6	Phase 5, once per behaviour	The same drawing six times: the nodes each declared behaviour runs on, and the SHARED nodes, in full colour; what it only passes through in half-tone; the rest of the island faded

Colour means the same thing on all five pages: it is the **Autoware stack** a node belongs to, taken from the legend of `ArchitectureMain.pdf`. What changes between phases is only how far in the picture is zoomed, and — from phase 3 onward — which ECU each box runs on.

Derived from `L4SafetyIsland_SoftwareDesign.md`, version 20261023, without modifying it.

Phase 1 — Autoware today, every node

The published Autoware node graph (`ArchitectureMain.pdf`), redrawn here in the same visual language as phases 2–5 so that the five read as one progression.
Nothing on this page is changed by the Safety Island — this is the picture the rest of the series modifies. Colour is the Autoware stack, from the original's own legend.



Stacks — the colours of `ArchitectureMain.pdf`

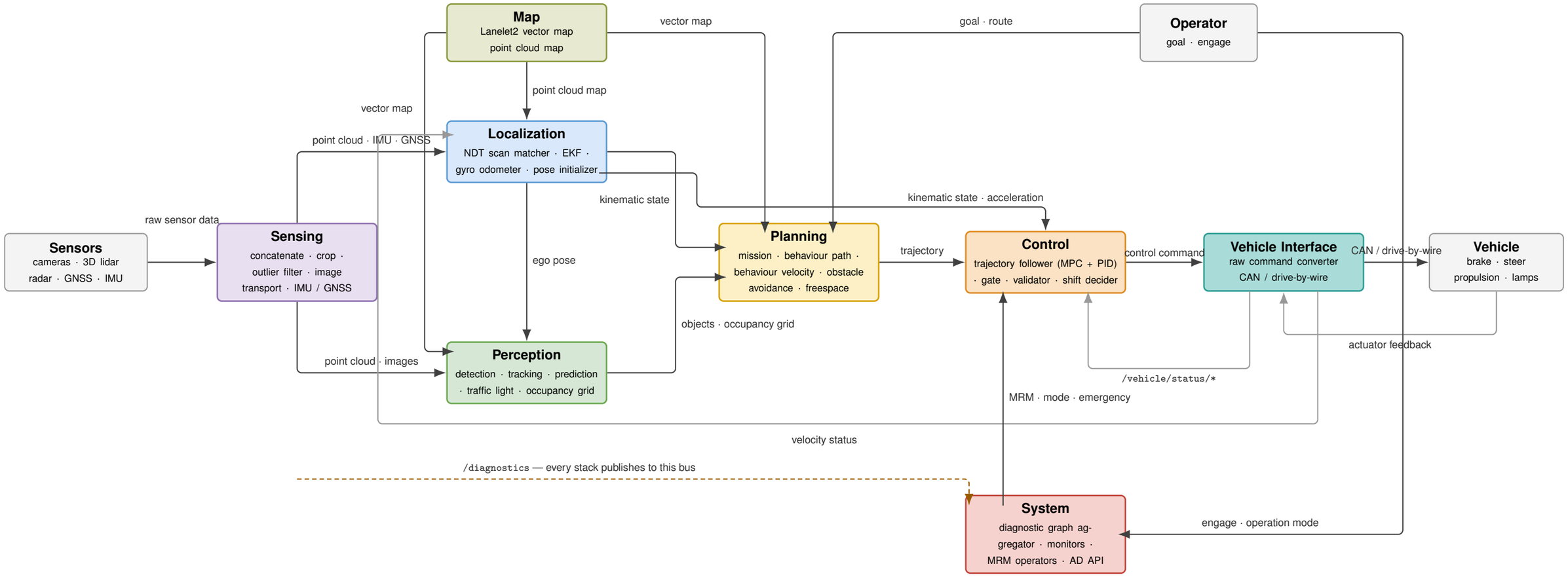
Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Map

This is the unmodified picture. Every node here stays exactly where it is until phase 3; phase 2 is this same graph with each stack merged into one box. Two condensations, for the page: the per-sensor lidar pre-processing chain is drawn once rather than once per sensor, and the two behaviour planners' module lists are given inside their box rather than as separate boxes.

Names are those of `ArchitectureMain.pdf`, which predates the `autoware_universe` renaming — `mpc_follower` is today's `mpc_lateral_controller`, `vehicle_cmd_gate` today's `control_command_gate`, and so on. The mapping is in `../nodediagram/L4SafetyIsland-NodeDiagram.md` §2.

Phase 2 — Autoware today, one box per stack

Every node of phase 1 merged into the stack it belongs to, and every link merged with it.
This is the picture the Safety Island changes; phase 3 shows the same boxes after the change.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Map · Planning

Control · System · Vehicle Interface · Vehicle / actor

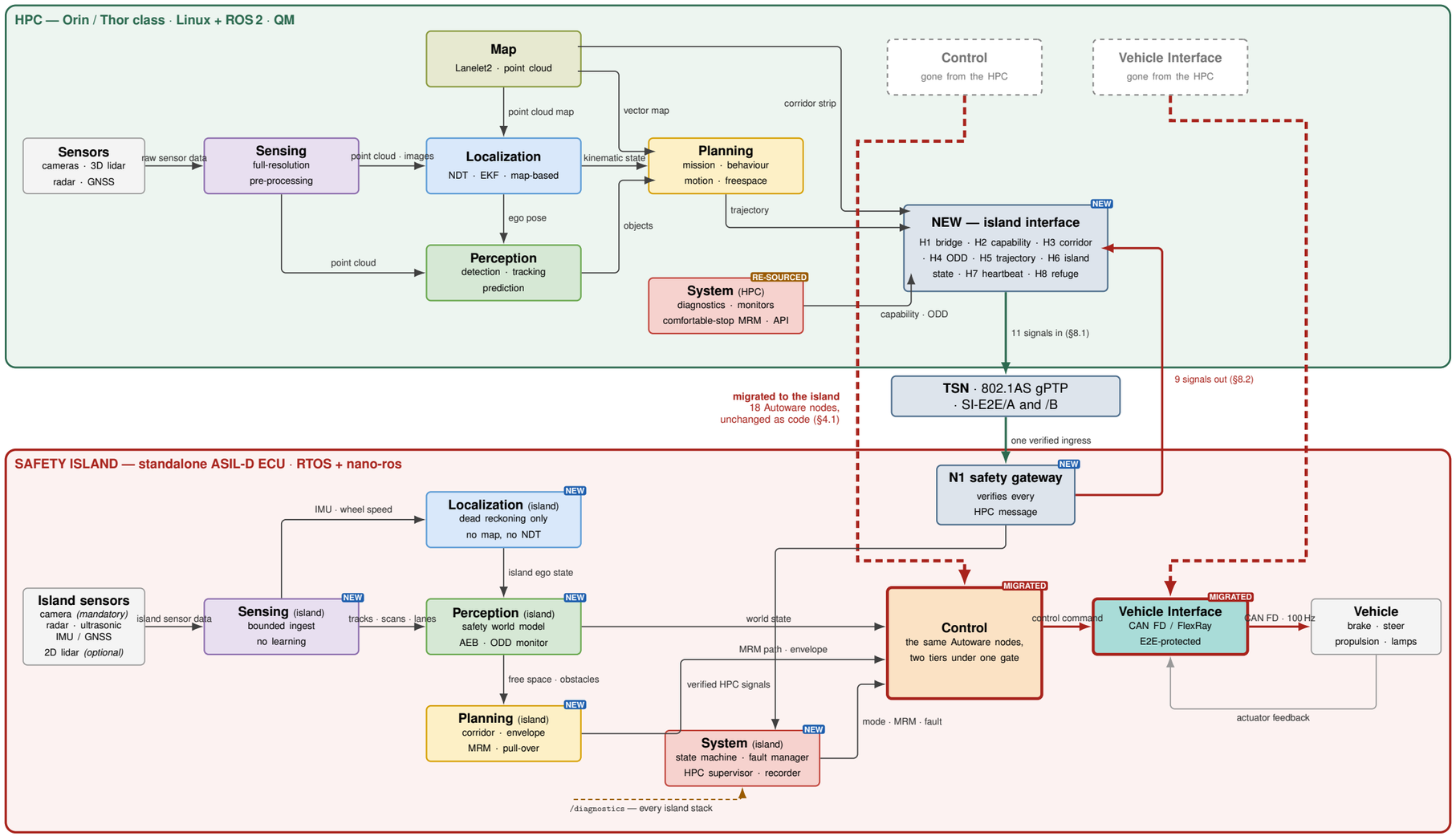
Merged links

- data, command or decision
- feedback from the vehicle
- diagnostics

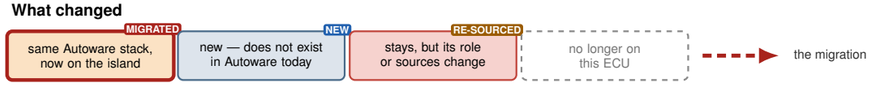
One box per stack, one arrow per merged link. The diagnostics bus is drawn once rather than as one stub per stack; every stack publishes to it. Phase 1 is the same graph with all of its nodes shown.

Phase 3 — the same picture, after the Safety Island

Same boxes and same colours as phase 2. **The Control stack and the Vehicle Interface move bodily to the island** — dashed red, from the ghost they leave behind. The island grows reduced versions of the other stacks (blue tag), the HPC gains an interface block, and one narrow link joins the two. Node-level detail is phase 4.



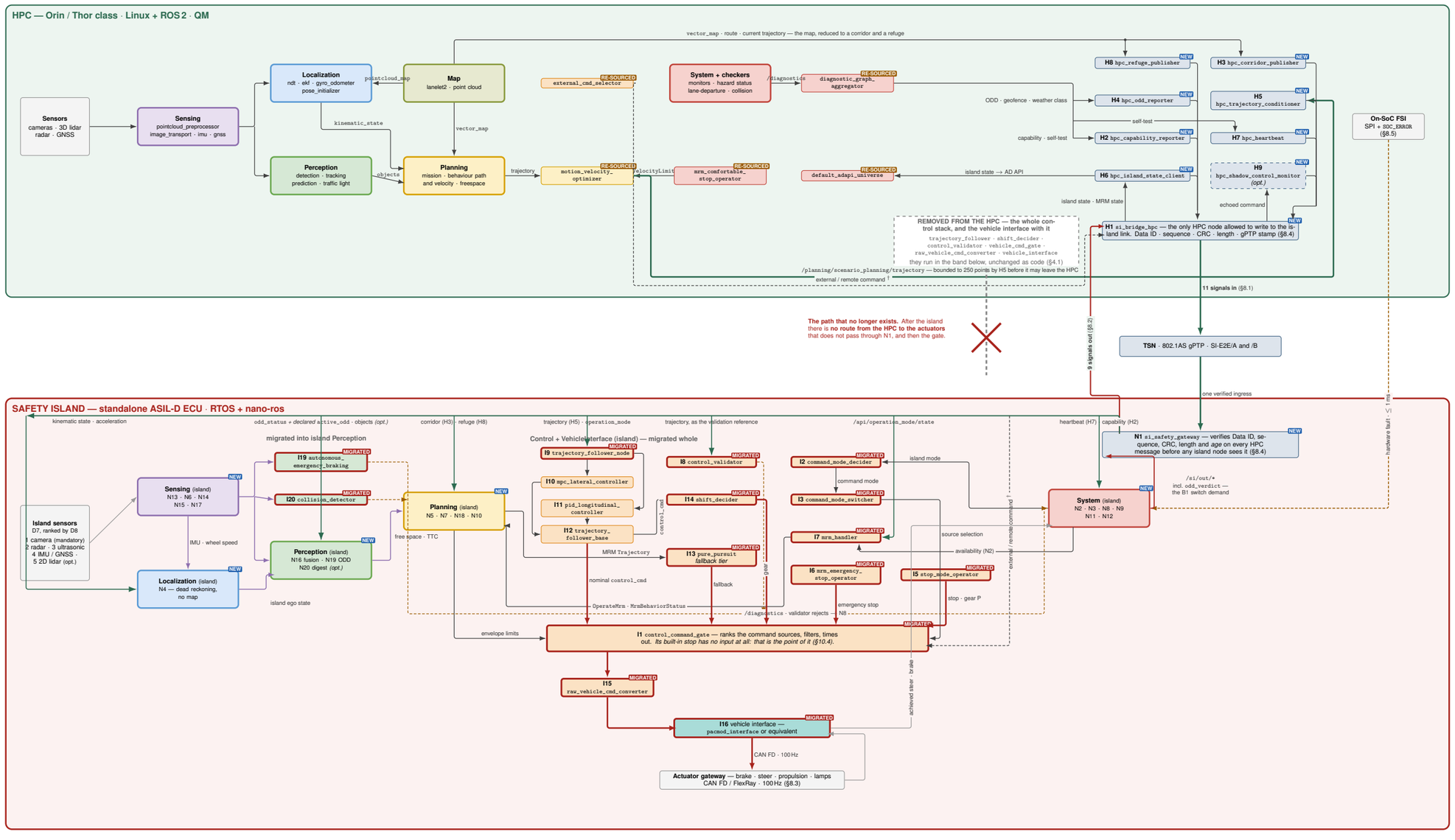
What changed



The island is not a monitor bolted on the side: it runs a *reduced Autoware*. Its localization is one node, its planning is four bounded map-free ones — and its control stack is not a reimplementation at all, it is Autoware's, relocated. Phase 4 opens what changed to node level.

Phase 4 — what the Safety Island changes, at node level

A **selective** zoom, between phase 3 (one box per stack) and phase 5 (the island, node by node).
On the HPC only the nodes that change are opened up — what was **removed**, what is **re-sourced**, and the **new** interface block. Every stack that the island leaves alone stays as one box, exactly as in phase 3.
On the island only the nodes migrated from the HPC are opened up. The island's own new stacks stay as one box each; phase 5 opens them.



How to read this phase

a whole stack, *unchanged* (phase 3) · a node, *migrated* to the island · a node, *new* · a node that *stays*, re-sourced · *removed* from this ECU

A heavy rule means a whole stack drawn as one box, at the phase-3 level of detail — the island changes nothing inside it. A thin rule means a single node, opened up because it changed. † the external / remote command reaches the gate (§5.2, §10.4) but has no row in the §8.1 interface table.

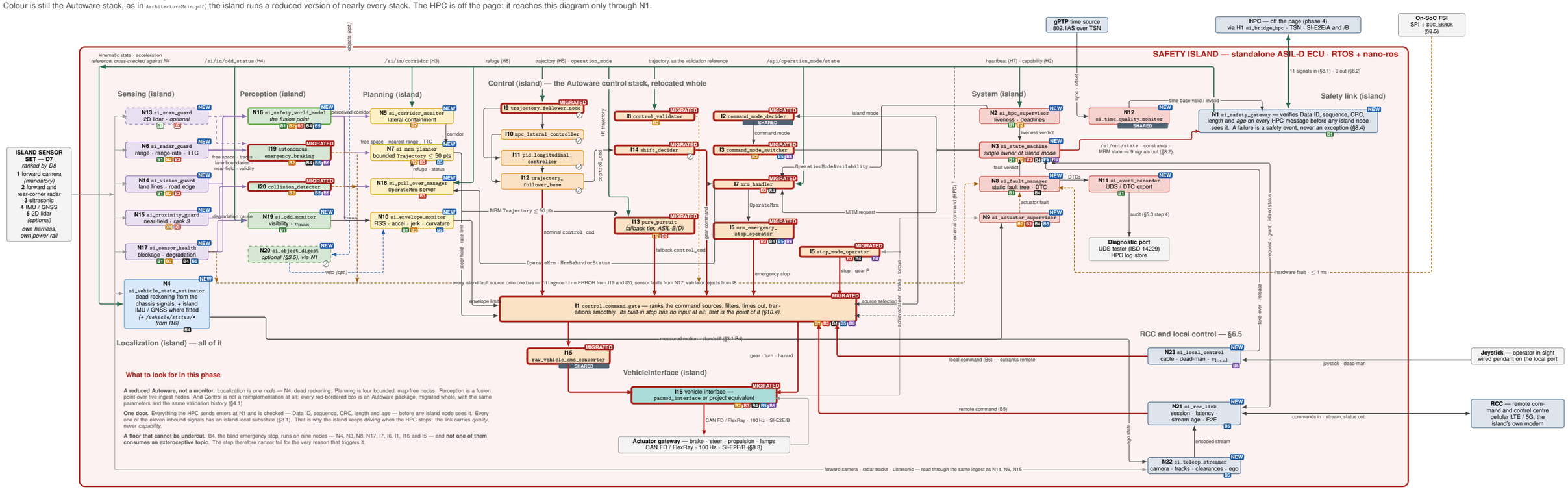
Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- the path that was removed

18 Autoware nodes migrate (§4.1) — every red-bordered box, plus I10–I12 inside I9 and I4, a header-only library with no topics. **19 new island nodes** sit inside the five blue-tagged island blocks; phase 5 opens them. The complete island edge catalogue is `../messageflow/`.

Phase 5 — the Safety Island alone, every node

The last zoom. All 37 island nodes — 18 migrated Autoware **I**-nodes and 19 new **N**-nodes — plus the optional N20, and N21–N23 of §6.5 for B5 and B6 (§6.6).
Colour is still the Autoware stack, as in `ArchitectureMain.pdf`; the island runs a reduced version of nearly every stack. The HPC is off the page: it reaches this diagram only through N1.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1 B2 B3 B4 B5 B6
SHARED
B2 B3
⊘

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

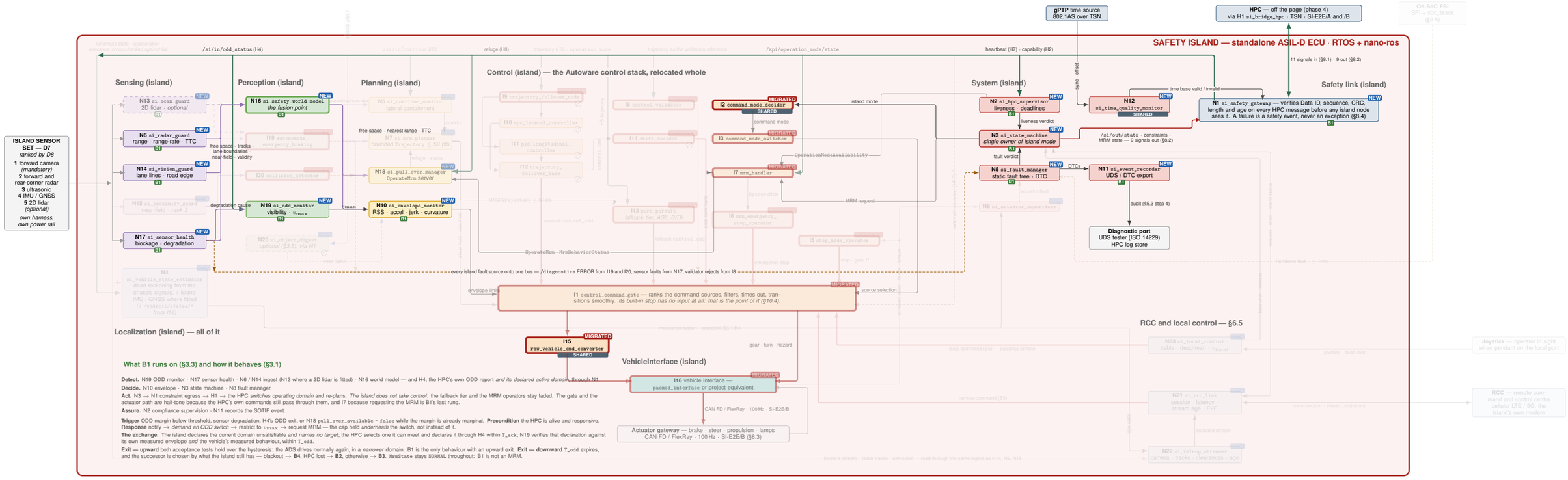
Messages

HPC-sourced, verified by N1
island sensor data
control command into the gate
fault · `/diagnostics`
actuator feedback
state · mode · decision
optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

Phase 5 · B1 — Degrade the ADS when the ODD cannot be held

When the operating envelope is no longer satisfied, require the ADS to move to an operating domain it *can* satisfy — progressively constraining it, and demanding that it constrain itself, until it does — before any authority is taken.
The phase-5 drawing, unchanged: what B1 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

In this view — B1, degrade the ADS

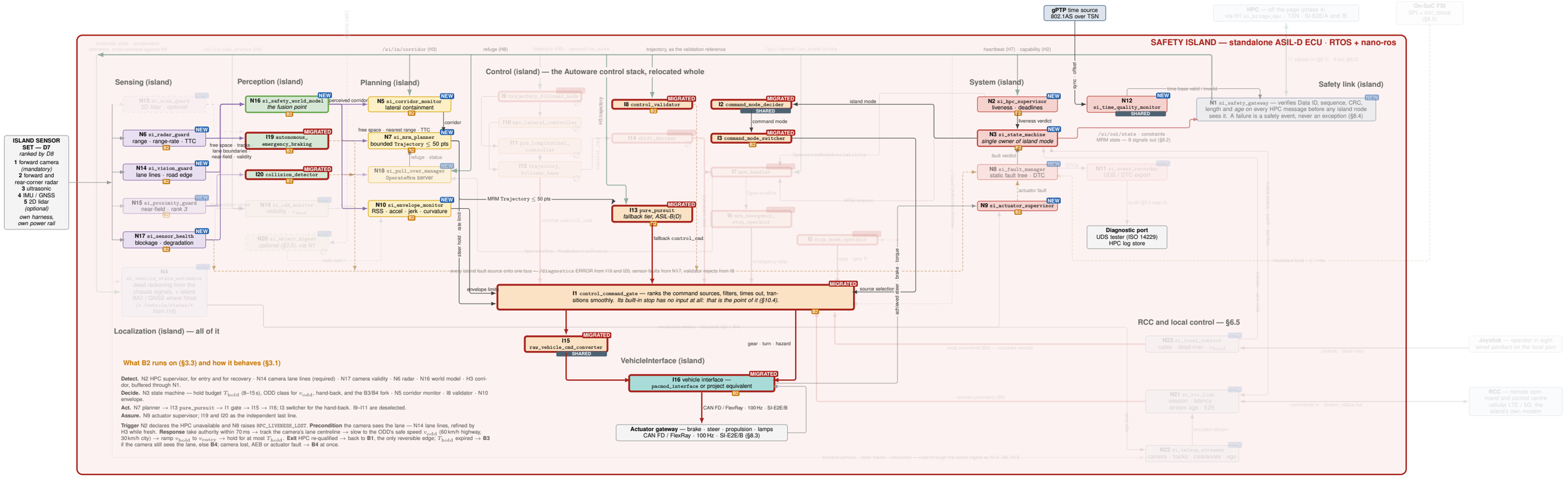
full colour — named for B1 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B1's path by §3.1, but not named in §3.3

faded — not used by B1. An edge is drawn at the weaker of its two ends. Only B1's own tag is drawn on each box.

Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded 8–15 s window, first down to the ODD's safe speed — while the camera still sees the lane.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour
B2
SHARED
B2

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B2, lane-keep hold

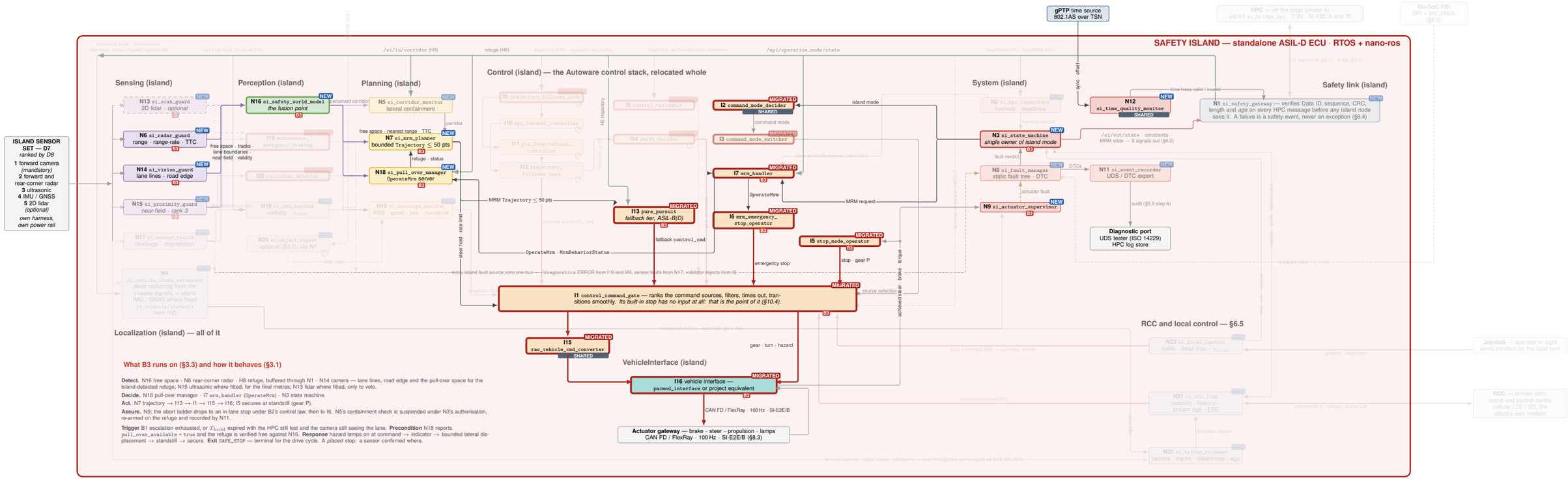
full colour — named for B2 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B2's path by §3.1, but not named in §3.3

faded — not used by B2. An edge is drawn at the weaker of its two ends. Only B2's own tag is drawn on each box.

Phase 5 · B3 — Pull over if the ODD continues to fail

When degradation does not restore the envelope within a declared deadline, bring the vehicle out of the running lane to a standstill in a refuge — nominated by the HPC, or found by the island's camera.
The phase-5 drawing, unchanged: what B3 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
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Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B1 · B2 · B3 · B4 · B5

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B3, pull over

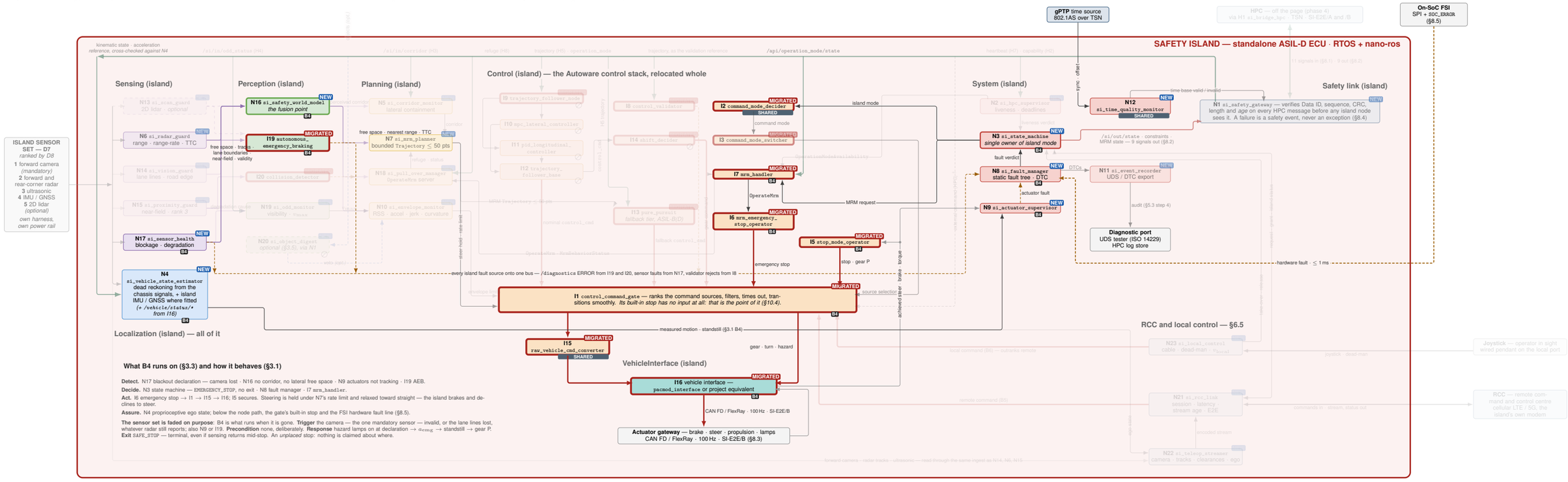
full colour — named for B3 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B3's path by §3.1, but not named in §3.3

faded — not used by B3. An edge is drawn at the weaker of its two ends. Only B3's own tag is drawn on each box.

Phase 5 · B4 — Emergency stop when the island is blind

When the island has lost the sensing that B2 and B3 both depend on — the camera — decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre.
The phase-5 drawing, unchanged: what B4 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

MIGRATED — the same Autoware node, migrated
NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour — B4 — SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · /diagnostics
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B4, blind emergency stop

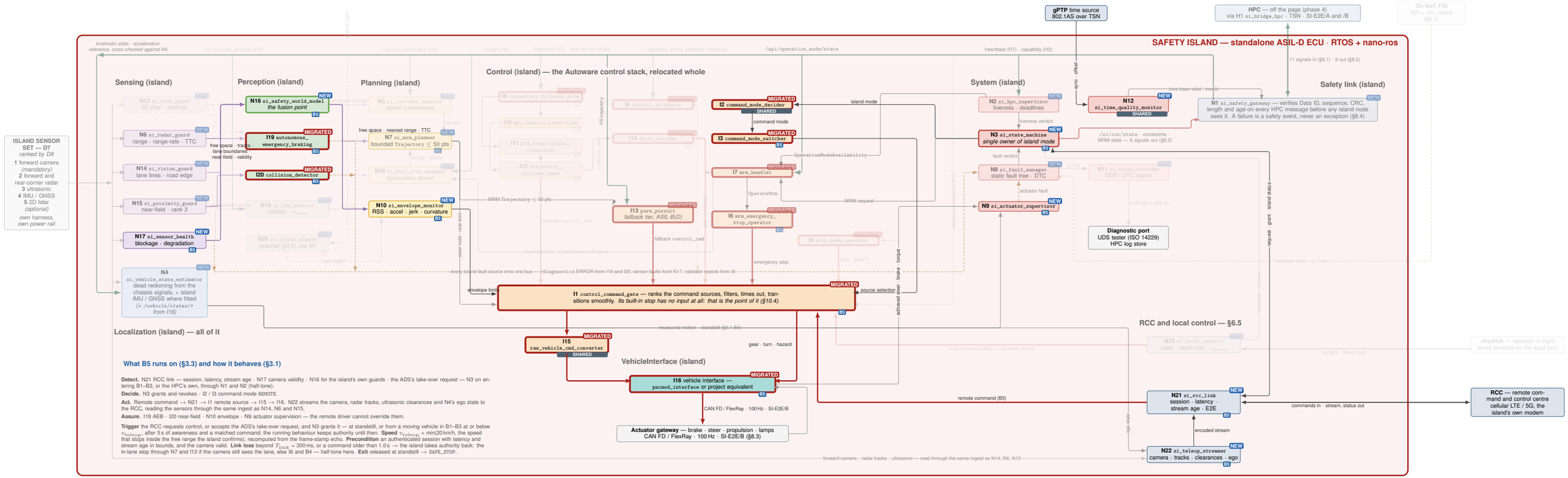
full colour — named for B4 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B4's path by §3.1, but not named in §3.3

faded — not used by B4. An edge is drawn at the weaker of its two ends. Only B4's own tag is drawn on each box.

Phase 5 · B5 — Tele-operation from the remote command centre

A remote driver at the RCC drives the vehicle over the island's own communication module, on the camera, radar and ultrasonic the island streams — while the island keeps its guards armed and stops the vehicle itself if the link fails.
The phase-5 drawing, unchanged: what B5 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Detailed design: §6.6 — from standstill, or taken over from a moving vehicle in B1–B3 on the ADS's request.*



Stacks — the colours of `ArchitectureMain.pdf`

Sensing · Localization · Perception · Planning · Control
System · VehicleInterface · Vehicle · Safety link

Where each node came from

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NEW — new — does not exist in Autoware

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B5 · SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · `/diagnostics`
- actuator feedback
- state · mode · decision
- optional (§3.5)

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In this view — B5, tele-operation from the remote command centre

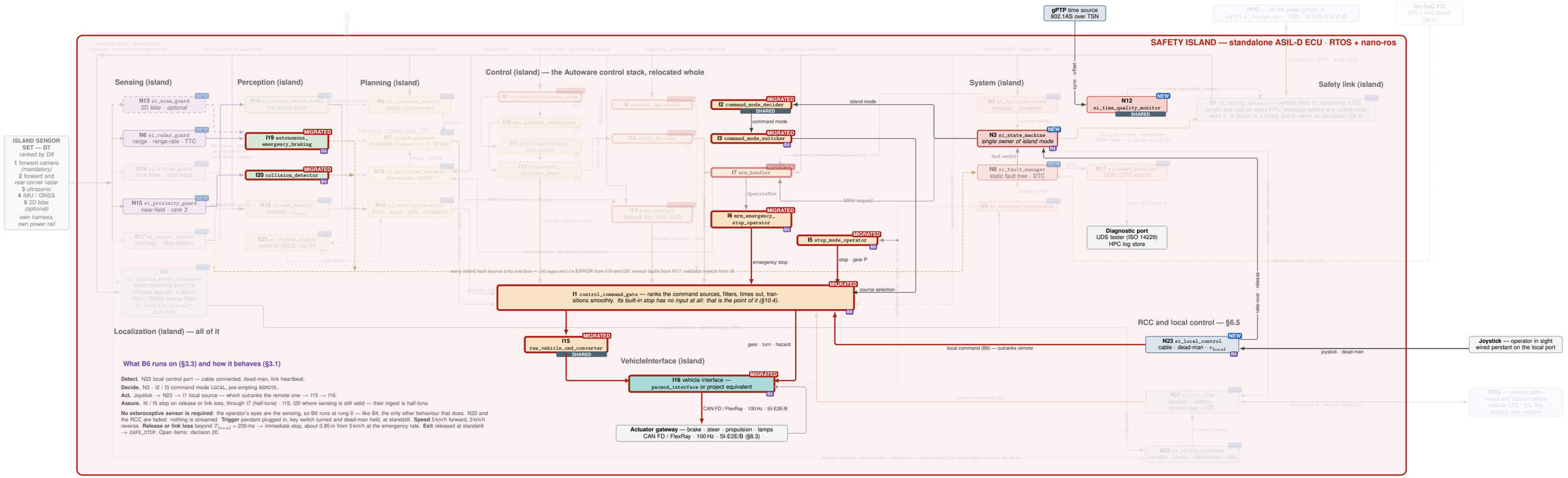
full colour — named for B5 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B5's path by §3.1, but not named in §3.3

faded — not used by B5. An edge is drawn at the weaker of its two ends. Only B5's own tag is drawn on each box.

Phase 5 · B6 — Manual joystick control without streamed sensing

An operator in sight of the vehicle drives it with a joystick on the island's local control port — by cable; a radio link is deferred to the next design — with no sensor stream; the island holds the speed to a creep and stops on release or on loss of the link.
The phase-5 drawing, unchanged: what B6 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Stage-1 detailed design: §6.6, entered from standstill.*



Stacks — the colours of `ArchitectureMain.pdf`

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System · VehicleInterface · Vehicle · Safety link

Where each node came from

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Behaviours each node enables (§3.3)

one fixed slot per behaviour — B6 — SHARED

B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- HPC-sourced, verified by N1
- island sensor data
- control command into the gate
- fault · /diagnostics
- actuator feedback
- state · mode · decision
- optional (§3.5)

Every edge on this page, with the design-document section it is taken from, is catalogued in ../messageflow/, which draws the same 37 nodes coloured by execution partition instead.

In this view — B6, manual joystick control without streamed sensing

full colour — named for B6 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)

half-tone — on B6's path by §3.1, but not named in §3.3

faded — not used by B6. An edge is drawn at the weaker of its two ends. Only B6's own tag is drawn on each box.